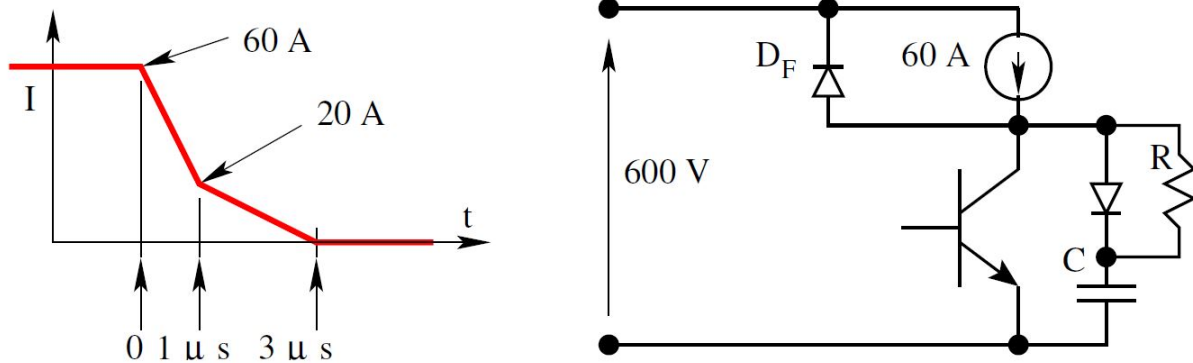
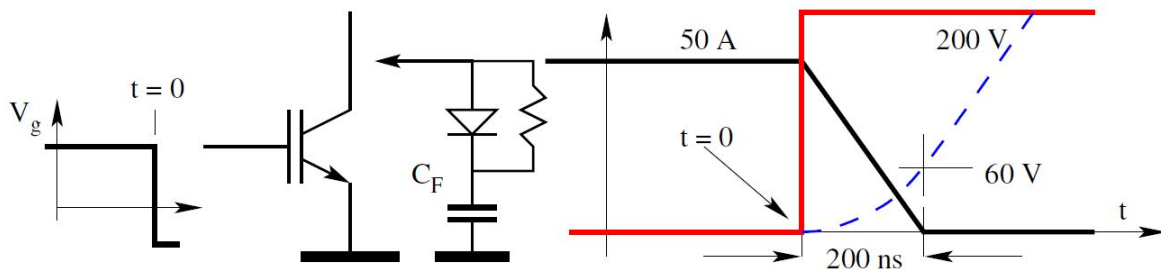


Assignment 3: Snubbers.

1. The following figure shows the current through the device when it is being switched off. The circuit voltage is 600 V. The load current is constant at 60 A. The freewheeling diode ensures that the load current has an alternate path when the switch is off. The RC circuit provides smooth recovery of the device voltage during turn-off. The value of C is $0.5 \mu\text{F}$. Assume the diodes to be ideal. (5)



- Sketch the capacitor current during and following turn-off.
 - Evaluate the voltage across the device during switch-off ($0 \mu\text{s}$ to $3 \mu\text{s}$).
 - Sketch the device voltage following switch-off ($3 \mu\text{s}$ to $6 \mu\text{s}$).
 - Following switch-off, evaluate the time taken for the capacitor C to get fully charged to 600 V so that the freewheeling diode D_F will conduct.
 - Evaluate the switching energy loss in the device during turn-off ($0 \mu\text{s}$ to $3 \mu\text{s}$).
2. Consider the following switching waveforms of a controlled switch in an application. The switch current on turn-off linearly drops to from 50 A in 200 ns. The switch voltage instantly rises to 200 V. (5)



- A. Evaluate the turn-off energy loss in the switch.
- B. A capacitor C_F in parallel with the switch can shape the recovery voltage (as shown by the dotted line of voltage recovery) of the switch to reduce the turn-off loss in the switch. Evaluate C_F in order to limit the voltage across the switch to 60 V at the end of current fall time as shown in the figure.
- C. Evaluate the turn-off energy loss in the switch for the modified turn-off transient.
- D. Evaluate the energy stored in the capacitor C_F at the end of the turn-off transient when the capacitor voltage has reached 200 V.